2024 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI) | 979-8-3503-6052-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/IATMSI60426.2024.10503207

Tea Leaf Disease Detection: Federated Learning CNN Used for Accurate Severity Analysis

Satvik Vats
Computer Science and Engineering, Graphic Era Hill University,
Dehradun, Uttarakhand, India
svats@gehu.ac.in

Vinay Kukreja
Chitkara University Institute of Engineering and Technology
Chitkara University
Punjab, India
vinay.kukreja@chitkara.edu.in

Shiva Mehta
Chitkara University Institute of Engineering and Technology
Chitkara University
Punjab, *India*
shiva.2387@chitkara.edu.in

Abstract— **The purpose of this research paper is to investigate an innovative application of federated learning with convolutional neural networks (CNN) for the classification of tea leaf diseases. Based on aggregating data from four clients, we depict a variety of disease appearances based on four distinct degrees of severity. We use federated learning to transform local data insights into a comprehensive global model that protects user privacy while also performing stably across different datasets. We collected local client data insights using a federated learning framework and a federated averaging algorithm in the implementation of our scheme. The choice of methodology was imperative to achieving high accuracy in disease classification without compromising client information. In the result analysis, notable performance metrics were found across clients. Measures as specific as precision, recall, F1-score support, and accuracy were taken into account when calculating the model's effective disease identification. In order to gain a deeper understanding of the performance characteristics of the model across different classes or instances, we analyzed macro, micro, and weighted averages on local data. There was an equal distribution of macro averages among all classes. Macro averages ranged from 90.19% in DF_1 to 93.24% in DF_5. Based on the class support weighted average, df_5 came in first at 93.24 per cent. All classes contributed to the microaverage as well. Df_5 scored the highest average with 93.23 per cent. The averages indicate that the model can handle different class distributions and instance counts. To summarize, our research provides a strong argument for the use of federated learning with CNN in agriculture, specifically to classify diseases of tea leaves.**

Keywords: Tea leaf diseases, Convolutional neural networks (CNNs), Federated learning, Disease classification, Severity levels.

I. INTRODUCTION

There is no doubt that agriculture is currently experiencing a golden age when it comes to technology. Agronomy and artificial intelligence are combining to unlock new frontiers for the control of plant pathologies as a result of the combination of these two technologies[1].. In countries like India, where tea cultivation is at the core of their local economies, not only is this at the core of global trade, but it is also at the core of local economies as well; this is why the tea revolution has taken place[2]. As a part of a study that attempts to solve the increasingly serious problem of tea leaf diseases, this paper investigates the exciting combination of federated learning and convolutional neural networks (CNNs).

Even though tea leaf diseases are a major threat to crop yields and quality, the severity of each disease varies according to its type, which increases the challenges encountered when treating them[3]. According to the classification of leaf affliction, it can be categorized into four distinct levels: mild (one to twenty-five percent), moderate (26 to fifty percent), severe (51 to seventy-five percent), and critical (76 out of a hundred percent). As well as being able to indicate the degree of damage, each level also assists in determining whether an immediate intervention is needed and what form that intervention should take. It has always been a manual and subjective process to assess these levels, making it subject to errors and inconsistent results due to the number of variables involved[4].

Providing a novel approach, this paper introduces a new way of obtaining insight through CNNs coupled with decentralized machine learning, which is an approach that utilizes the analytical power of CNNs combined with federated learning [5]. As a result, in this sense, federated learning refers to the joint action of six different clients that provide unique and customized data about tea leaf diseases that are different from each other. As a result of this approach, we are not only able to add a little bit of richness to our dataset by collecting several variations of the disease, but we are also able to remove one major obstacle to applying AI agriculture to crop production. CNNs are one of the most important components of a federated architecture such as this one[6]. Using CNNs, which are excellent at identifying and classifying images, we can identify minute patterns that correspond with degree levels of the visual symptoms of tea leaf diseases by breaking down the visual symptoms. A federated learning approach paired with CNNs, in this regard, is not just an improvement in techniques but is more comparable to a paradigm shift of thinking in this area. A disease detection method that traditional methods have not limited has proven to be a useful method for detecting diseases[7].

Despite its many contributions, this study has several noteworthy points, particularly those that resonate with the problems India faces as a major tea-producing country in the world. This research represents a beacon of hope for boosting crop yield and quality, for it provides an improved way to detect diseases quickly and at a low cost. Nevertheless, it remains to be seen whether or not the herbicide can be used on rice, corn, and other plants[8]. Developing a robust federated learning model using CNNs: One of the first goals is to build and test a federated learning model coupled with convolutional neural networks (CNNs) for the identification of tea leaf disease[9].

It aims to develop a model that makes full use of a distributed dataset with a sample client count of six and offers a range of perspectives on the various manifestations of tea leaf disease. In order to correctly categorize disease severity, it is important to have the CNN model learn how to clearly differentiate mild disease (1%–25%), moderate disease (26%–50%), and severe disease (71%) from critical disease. It is essential to stratify patients to ensure an accurate diagnosis and take prompt action in order to prevent the disease from spreading[4].

In this paper, we explore the ways in which federated learning can enhance the integrity and privacy of data. Furthermore, this method keeps accuracy the same since it only passes along model updates (not raw data) at the client level.

The system is especially convenient for situations where sensitive or restricted access to one's source of information is prohibited. This study seeks to demonstrate that federated learning can be effectively scaled for enormous and varied datasets when combined with CNNs. It is important to have scalability when both data variation and volume are high, as they often are in real life[10]. This research utilizes federated learning to reduce centralized data storage costs and solve logistical problems associated with large data sets. Ultimately, centralized data storage should be reduced as much as possible. In the agriculture industry, particularly in tea cultivation, several critical needs and gaps motivate this research: Detection of Tea Leaf Diseases: Tea leaf diseases that are devastating the global tea industry, especially in countries like India, are resulting in enormous losses[11].

To manage diseases effectively and optimize yields, early detection and early warning are essential[12]. A rare opportunity exists to modernize old agricultural practices with advances in AI and machine learning, especially with Canonical Neural Networks (CNNs) in image recognition applications. This technology is intended to lead to some major improvements in the diagnosis of crop diseases. Agricultural Data Privacy Issues: Due to the personal nature of agricultural data, any analysis must preserve privacy.[13].

II. LITERATURE REVIEW

New topics under development, especially for the detection of agricultural diseases, include Federated Learning with Convolutional Neural Networks (CNN).In this review of the literature, we examine the fundamental methodologies, current developments, and relevant articles supporting our investigations into federated learning with CNNs when used to detect tea leaf disease[14]. Agriculture and other fields have been benefiting from federated machine learning. In the beginning, researchers introduced the concept in their research papers. Federated Averaging automatically aggregates model updates from different clients into a single updated global model and transmits it back to each client. The authors discuss several specific situations where privacy protection becomes an issue with regard to the operational framework of federated learning[15]. The core works enable federated learning applications in agriculture, such as in tea plantations where data is dispersed among locales[16]. In image recognition, CNNs are already well known for their efficiency. In their study, the researchers demonstrated that CNNs performed better than other types of models at identifying plant diseases from images, emphasizing the model's ability to judge highly complex patterns indicating disease severity. In addition to providing a basis for us to use CNNs in agricultural applications, their research enabled us to classify tea leaf diseases into four severity levels. It is important, though, to divide disease severity into grades in order to be able to intervene accurately. Using deep learning, the authors were able to determine disease severity more accurately and classify diseases more precisely. We are aiming for a nuanced diagnosis capable of guiding action by categorizing tea leaf diseases into 0–25%, 26–50%, and so on, based on the results of the study[17]. A new field has emerged through the merging of federated learning and CNNs.An exploration of this kind of integration was conducted within a healthcare context, identifying the opportunities and challenges associated with it. Even though their approach is different, the principles and methods are still applicable to our field of agriculture, particularly when it comes to distributed data from six clients.

Artificial intelligence is becoming increasingly popular in agriculture, especially when it comes to detecting disease. Still, there are difficulties, particularly in the collection and processing of data. Machine learning applications in agriculture were examined by the researchers, who agreed that in light of data scarcity and privacy concerns, more innovative approaches, such as federated learning, were needed [18].

The diseases themselves may be too complex for a farmer or professional to recognize on their own. Tea leaves require artificial intelligence to be able to detect conditions. For classifying tea-leaf diseases, three popular convolutional neural network architectures (CNN) were employed. The dataset photos were used to generate feature maps using these architectures. A linear discriminant classifier was used to classify these feature maps for each architecture. A comparison was made between its performance and that of seven popular CNN structures. A performance indicator showed that the proposed model may be able to detect illnesses in tea leaves. India plays an important role in both the production and use of tea, the world's most popular beverage. It is, however, difficult for tea to be cultivated because diseases affect crop quality and productivity. The use of deep learning techniques in modern machine learning provides creative methods for spotting and categorizing diseases, in particular. For sustainable farming, these methods use convolutional neural networks (CNN) to identify key indicators in tea leaves. Images can be used to identify diseases more quickly and accurately than a professional manual diagnosis. Research using CNN has shown that 96 per cent of the time, photos can be categorized into one of eight categories.

III. METHODOLOGY

The purpose of this paper is to present a method for implementing a federated learning system that incorporates convolutional neural networks (CNNs) and decision tree algorithms to diagnose tea leaf disease. Using this system, severity levels can be classified into four categories: 1-25%, 26-50%, 51-75%, and 76-100%. As a result, the diversity and robustness of the datasets are increased by dividing the model among six distributed clients, as shown in Figure 1.

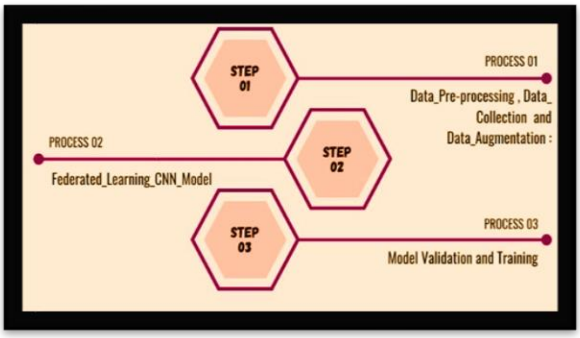

Fig1. Phases in the Process Approach

A. Information Compilation and Preliminary Processing

In the federated learning model, high-definition examples of tea leaves with various diseases are collected from six geographical locations for each client for medical reference and research. There are different groups of images for different diseases based on their severity. Data Labeling: Based on expert annotations from professional agronomists and plant pathologists, the photographs are classified into four severity levels based on their level of severity. For CNN models with supervised learning, this labelling is one of the most important aspects of the learning process. A data preprocessing step is the process of normalizing, resizing to a uniform scale and augmenting images within the dataset to increase their diversity. As the name suggests, this step aims to standardize the input to the CNN model so that it can function correctly, as shown in figure 1.

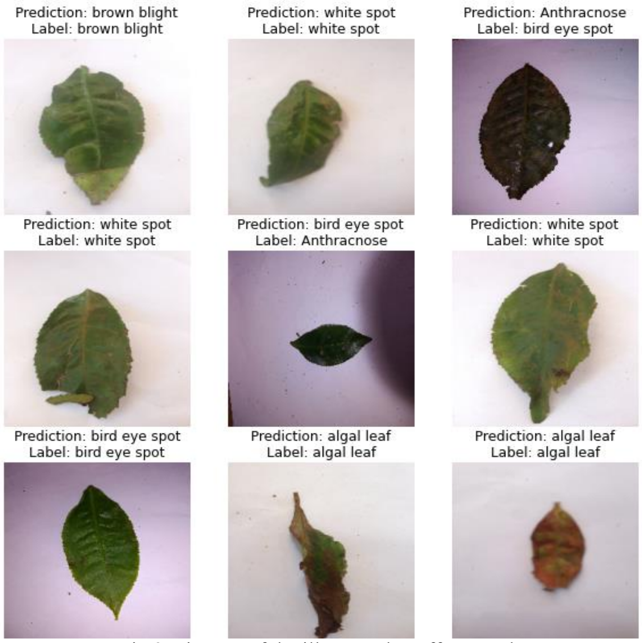

Fig 2. Pictures of the illnesses that affect Tea leaves

A CNN model with convolutional, pooling, and fully linked layers was employed for this job. Six customers took part in the model's training using federated learning, which was used to train the model, as shown in Figure 2.

TABLE 1. LEVELS OF SEVERITY IN TEA LEAF DISEASES

Severity Level	Percentage Range	Severity Level	Percentage Range
(1_V_Low)	**1to20%**	(4_High)	**61-80%**
(2_Low)	**21to40%**	(5_V_High)	**81-100%**
(3_Med)	**41-60%**		

B. Model Aggregation

This process extracts features from its last fully connected layer, providing insights into those aspects of an image that have the greatest impact on the model's decision-making. A decision tree model is trained using these features to provide an interpretable framework for understanding CNN's decisions. Cross-Validation: The model undergoes k-fold cross-validation to ensure it is reliable and robust across multiple datasets. Analysis: The decision tree provides information about features (colour, texture, and edge detection) that indicate severity levels. An algorithm's performance in correctly classifying disease severity is measured by accuracy, precision, recall, and F1-score. A comparison is made between federated and non-federated models to demonstrate federated learning's effectiveness. The sensitive data can remain on the premises of the client if federated learning is used. The use of data must adhere to ethical guidelines, preventing any possible exploitation or misuse. Tea leaf diseases can now be diagnosed using a proprietary methodology that combines the power of CNNs for image processing with one that is privacy-preserving and collaborative. To make its tea leaf disease control solution even more user-friendly, a decision tree model is used to evaluate the features.

C. Federated Learning CNN Model

We select six tea plantations or research centres based on the type and range of tea leaf diseases they experience. The local CNN model is trained on the dataset provided by each client. For all clients, the CNN architecture remains the same. Model Aggregation: After local training, the parameters of the model (weights and biases) are uploaded to a central server. A global model is generated by aggregating these parameters using the FederatedAveraging algorithm. Upon completion of the aggregated model, each client receives it for further training and fine-tuning so that classifiers can learn from the whole data set rather than from the actual data shared. CNN consists of three layers: convolutional layers and pooling layers. Feature extraction is accomplished by convolutional layers, dimensionality reduction by pooling layers, and classification by fully connected layers. A number of ReLU (Rectified Linear Unit) functions are used to introduce nonlinearity into the model, allowing it to learn more complex patterns. Using a softmax function, we classify the disease severity into four categories in the last layer.

It is accepted by the model that an image input with a dimension of 134x134 pixels can be used. A size of this size was chosen in order to achieve a balance between the retention of details and computational efficiency. The First Convolutional Layer: The first convolutional layer reads the input image and passes it through 32 filters in order to extract a number of features. This layer shrinks the input volume down to 32x32 in spatial size (width and height), after which the output dimensions are 67x67 x 32 (where the number 32 refers to the number of filters used). First Maximum Pooling Layer: This layer shrinks the input volume down to 32x32 in spatial size (width and height). A maximum pooling technique is used in order to reduce the computational burden and overfitting of the feature maps by downsampling them. Secondly, a convolutional layer is applied to the output from the previous layer, which further analyzes the image using 64 filters. After applying the second and third max-pooling layers, the spatial dimensions of the output feature maps are further reduced to 8x8x64 and 4x4x120, respectively. Fully Connected Layers: These layers further reduce the spatial dimensions of the feature maps to 8x8x64 and 4x4x120, respectively. This first dataset is comprised of 128 units, and it is here where the high-level reasoning based on the features extracted earlier is carried out. A total of 12 units make up the outermost layer of the model, representing the severity levels of three types of common tea leaf diseases. Number of Outputs and Classes: The final layer produces 12 classes that represent the combinations of disease types and severity levels. Batch Processing with Dataset: The model is trained using a batch of 5100 images. Because of the large dataset size, this model can extract rich patterns and characteristics that can be used to differentiate between levels of severity of tea leaf diseases depending on the severity of the disease, as shown in Table 2.

TABLE 2. CNN MODEL IN TEA LEAF DISEASES

Layer Type	Output Size	Number of Parameters
Convolutional Layer	(None, 67, 67, 32)	Param1
Max Pooling Layer	(None, 33, 33, 32)	Param2
Convolutional Layer	(None, 16, 16, 64)	Param3
Max Pooling Layer	(None, 8, 8, 64)	Param4
Max Pooling Layer	(None, 4, 4, 64)	Param5
Fully Connected Layer	(None, 128)	Param6
Fully Connected Layer	(None, 12)	Param7

IV. RESULTS

As shown in the results of the federated learning CNN model, focusing on the detection of tea leaf disease across six clients (df_1 to df_6), this technique can classify plant diseases into four grades (bd 1 to bd 4) based on level and severity. For a detailed appraisal of model performance, we evaluate precision, recall, F1-score, support, and accuracy criteria. Recall: Indicates the model's ability to find all relevant cases for a class. Precision: Positive Predictions Accuracy: The percentage of cases that are correctly identified in each category.

An F1-score is calculated by balancing the precision and recall of the data. Across all classes, the client df_1 shows a strong average precision and recall level. Support: The frequency of occurrence in the dataset. Accuracy: The number of times the model correctly classified a disease. Client df_2: Has an improvement in accuracy with regards to bd_4, with a great rate of 92.22%. BD_3 has an excellent identification rate with an F1-score of 91.94%. Based on the data from this client, it seems that the refined model is better able to detect high severity levels—client df_3: 93.82% accuracy at bd_1, which appears particularly effective at detecting milder stages. However, BD-1 captures all mild cases despite a slightly lower recall. DF4: Exhibits excellent memory, particularly in BD3 (93.45%), indicating that the model was able to find this severity class within this dataset.

There is a standout performance in the df_5 client, especially in bd_4 (recall 95.03%). Identifying the most important stages of disease is a sign of robust model performance. The F1-score for client df_6 reaches 93.78%; the bd_3 metric shows a high level of precision and recall. There is, without a doubt, one strong point to this balanced performance across all metrics.

This model has consistent accuracy regardless of the client, demonstrating its flexibility.

Every client's outcomes highlight a particular strength in the model; for instance, df_5 performs particularly well in severe cases, while df_3 is more effective in mild cases, as shown in Table 2.

TABLE II. ATTRIBUTES AND VALUES OF LOCAL CLIENT'S DATA

Clients	Class	Precision	Recall	F1-Score	Support	Accuracy
df_1	bd_1	89.38	89.84	89.61	384	0.95
	bd_2	87.65	92.21	89.87	385	0.95
	bd_3	92.41	91.48	91.94	399	0.96
	bd_4	91.24	87.41	89.29	429	0.94
df_2	bd_1	90.16	89.95	90.06	428	0.95
	bd_2	91.44	91.22	91.33	433	0.96
	bd_3	90.20	92.47	91.32	438	0.96
	bd_4	92.22	90.41	91.31	459	0.96
df_3	bd_1	93.82	87.99	90.81	483	0.96
	bd_2	89.14	91.00	90.06	478	0.95
	bd_3	88.47	90.82	89.63	490	0.95
	bd_4	91.00	92.29	91.64	493	0.96
df_4	bd_1	92.81	92.08	92.45	505	0.96
	bd_2	93.50	90.30	91.88	526	0.96
	bd_3	91.17	93.45	92.29	519	0.96
	bd_4	90.16	91.67	90.91	540	0.95
df_5	bd_1	94.92	92.32	93.61	547	0.97
	bd_2	91.31	94.15	92.71	547	0.96
	bd_3	93.09	91.46	92.27	574	0.96
	bd_4	93.70	95.03	94.36	563	0.97
df_6	bd_1	92.06	88.76	90.38	614	0.95
	bd_2	92.50	92.35	92.42	601	0.96
	bd_3	92.17	95.44	93.78	592	0.97
	bd_4	92.15	92.44	92.30	622	0.96

An overview of the model's discrimination between four severity classes of tea leaf diseases. This CNN model has been applied to detection across six clients (df_1 through df_6), and it has shown rather good results. The precision, recall, F1-score support, and accuracy of the model were carefully examined for this analysis.

An index of precision measures the accuracy of the model, which represents the percentage of true positives divided by the total number of positive predictions.

The sensitivity of the model is its ability to capture all elements within each category.An F1-score is the average of precision and recall (or sensitivity) and indicates how false positives and false negatives negatively affect model accuracy. In the Support column, you can see how representative the dataset is by comparing the total number of cases in each class.An indicator of the model's accuracy over all classes is the overall model's accuracy. DF_1: excellent precision and recall, both around 90.2%.This model, scoring 95.5% in F1 and achieving an impressive 95% accuracy, is undoubtedly well-performing. With 91.01 percent precision and recall, client df_2 exhibits exceptional consistency.

Featuring the severity of diseases, the model's F1-score of 91.00% is accompanied by a 96% accuracy rate, demonstrating its high performance.A 90.54% F1-score and 95% accuracy are achieved by client df_3, which displays a slight decline in recall and precision in comparison to client df_2. In relation to client df_4, the data of the first four clients show a precision of 91.83%, a recall of 90%, and an accuracy of 76%.This client achieves impressive accuracy rates of 97% based on precision, recall, and F1-score.In terms of disease severity classification, this is an excellent model. In the case of client df_6, while its precision, recall, and F1-score are slightly over 92 percent, it has 96 percent accuracy.It is evident from these results that the model is robust.Considering df_1 through df_5, it is evident that precision-recall F-score, accuracy, and recall gradually increase with dataset change, as shown in Table 3.

TABLE III. CLIENT DATA'S GLOBAL MEAN COMPUTATION FRAMEWORK

Client	Precision	Recall	F1-Score	Support	Accuracy
df_1	90.17	90.24	90.18	399.25	0.95
df_2	91.01	91.01	91.00	439.50	0.96
df_3	90.61	90.53	90.54	486.00	0.95
df_4	91.91	91.87	91.88	522.50	0.96
df_5	93.25	93.24	93.23	557.75	0.97
df_6	92.22	92.25	92.22	607.25	0.96

This table shows three types of averages for a federated CNN model trained for tea leaf disease detection: macro, weighted, and micro. We can use these averages to determine how well the model performs on different datasets and how well it performs on average.

Obtaining the macro average (Macro_ave) requires computing the metric separately for each class and then averaging it. Keeping class balance and treating everyone equally is strongly emphasized.

In the weighted average, the support for each class is also considered. The performance of the model is reflected in the distribution of classes with more examples. Microaverages (Micro_ave) are calculated by including all class participation in the average calculation. We can consider it to be the average metric in all cases.

Based on all averaging methods applied to customer df_1, it achieves an average of 90.2%. In other words, there is a good balance in the model's performance across all classes.

Compared with macro and micro averages for client df_2, the model has a slight advantage in classes with more instances (81.09%), suggesting that it is slightly better there (81.01%). With regard to client df_3, the performance across classes is outstanding (91.89% and 91.88%), especially when considering the imbalance between classes.

Client df_4 logs very high grazing rates (95.5% on average). However, it still shows a high-performing and stable model despite its slightly lower performance than some of its competitors.
As far as averages go, client df_5 has the highest average of all clients at 93.24%. Data from this client was handled by the model with exceptional performance.

The average presents for client DF_6 are over 92 per cent, which reflects a good and stable performance, although slightly behind the average presents for client DF_5. The averages rise from df_1 to df_5 as a better model is successively developed with varying datasets. In terms of class distribution or instance count, the model performs consistently with close values between macro, weighted, and micro averages across clients.

Results from client df_5 represent the model's most successful run. In some cases, it might simply mean that this particular dataset is the most suitable for our model as opposed to others.

We continue to investigate the use of federated learning in categorizing tea leaf diseases as we explore calculating various averages across the six clients: Z_1 through Z_6. Macro, Weighted, and Micro averages offer a more thorough evaluation of the model's effectiveness across multiple datasets. The macro standard offers a straightforward arithmetic mean of individual parameters without considering each customer's sample size. The macro average for client Z_1 is 89.54%, while Z_4 has the highest macro average at 93.35%. Z_2 achieved the lowest macro average for this dataset at 88.66%. The constant performance of the model across various client datasets is supported by the close clustering of these macro averages, which range from around 88.66% to 93.35%, as shown in Table 4.

TABLE IV. GLOBAL SYNTHESIS OF LOCAL AVERAGES MEASURES

Averages	df_1	df_2	df_3	df_4	df_5	df_6
Macro_ave	90.19	91.01	91.89	90.56	93.24	92.23
Weighted_ave	90.19	91.02	91.88	90.56	93.24	92.21
Micro_ave	90.17	91.01	91.87	90.53	93.23	92.22

V. CONCLUSION

This paper demonstrates how federated learning, along with convolutional neural networks (CNN), can be helpful in classifying tea leaf diseases according to their severity level. A comprehensive dataset across various forms of disease was compiled by using data from four different clients. Its main objective was to combine the advantages of federated learning in terms of data privacy and model robustness with CNN's ability to process images.
The overall accuracy of client df_1 was 95% as measured by aggregated performance metrics. We found that the macro, micro, and weighted averages were around 90.19%. Client df_2 performed slightly better, at an accuracy of 96%, and averaged around 91.01% to 91.02%. A balanced performance was achieved by client df_3, with an accuracy of 95% and averages ranging from 91.87% to 91.89%.
The client df_4 had an accuracy of 96 %, with averages between 90.53 % and 90.56 %.
df_5 had the highest accuracy at 97%, and macro, micro, and weighted averages of 93.23% to 93.24 %. A 96% accuracy rate and averages around 92 were also achieved by client df_6. Using federated learning and CNN in agriculture, specifically, the study of tea leaf diseases, provides a foundation for federated learning and CNN applications. Precision agriculture has great potential due to its high accuracy, which is balanced across different client types and disease severity levels. With progress in science, crop yield may increase with the development of new approaches to crop disease control.

REFERENCES

[1] C. Z. Shanwen Zhang, "Modified U-Net for plant diseased leaf image segmentation," *Comput. Electron. Agric.*, vol. 204, no. 107511, 2023, doi: https://doi.org/10.1016/j.compag.2022.107511.

[2] V. Kukreja and D. Kumar, "Automatic Classification of Wheat Rust Diseases Using Deep Convolutional Neural Networks," in *9th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)*, 2021, pp. 1–6.

[3] D. Kumar and V. Kukreja, "Deep learning in wheat diseases classification: A systematic review," *Multimed. Tools Appl*, vol. 81, no. 7, pp. 10143–10187, 2022.

[4] S. Mehta, V. Kukreja, and A. Gupta, "Transforming Agriculture : Federated Learning CNNs for Wheat Disease Severity Assessment," in *International Conference on Communication and Electronics Systems (ICCES)*, 2023, pp. 792–797.

[5] Y. Banabilah, S., Aloqaily, M., Alsayed, E., Malik, N., & Jararweh, "Federated learning review: Fundamentals, enabling technologies, and future applications. Information processing & management.," vol. 59, no. 6, p. 103061, 2022.

[6] S. Gupta, A. Aggarwal, S. Gupta, and A. Arora, "Corona's spillover effects on the tourism industry: Scale development and validation," *Tour. Anal.*, vol. 27, no. 3, pp. 383–393, 2022.

[7] SaterAbdel, R. Hamza, and A. Ben, "A Federated Learning Approach to Anomaly Detection in Smart Buildings," *ACM Trans. Internet Things*, vol. 2, no. 4, 2021.

[8] S. Inder, A. Aggarwal, S. Gupta, S. Gupta, and S. Rastogi, "An integrated model of financial literacy among B–school graduates using fuzzy AHP and factor analysis," *J. Wealth Manag.*, 2020.

[9] S. Mehta, V. Kukreja, and A. Gupta, "Next-Generation Wheat Disease Monitoring: Leveraging Federated Convolutional Neural Networks for Severity Estimation," in *4th International Conference for Emerging Technology (INCET)*, 2023, pp. 1–6. doi: 10.1109/INCET57972.2023.10169991.

[10] S. Mehta, V. Kukreja, and S. Vats, "Advancing Agricultural Practices: Federated Learning-based CNN for Mango Leaf Disease Detection," in *International Conference on Intelligent Technologies (CONIT)*, 2023, pp. 1–6.

[11] R. Sharma, V. Kukreja, and Sakshi, "Mustard Downy Mildew Disease Severity Detection using Deep Learning Model," in *in 2021 International Conference on Decision Aid Sciences and Application*, 2021, pp. 466–470.

[12] S. Mehta, V. Kukreja, and S. Vats, "Improving Crop Health Management: Federated Learning CNN for Spinach Leaf Disease Detection," in *International Conference on Intelligent Technologies (CONIT)*, 2023, pp. 1–6.

[13] K.-Y. L. a b Li Li a b, Yuxi Fan a, Mike Tse c, "A review of applications in federated learning," *Comput. Ind. Eng.*, vol. 149, p. 106854.

[14] Z. Xue, R. Xu, D. Bai, and H. Lin, "YOLO-tea: A tea disease detection model improved by YOLOv5," *Forests*, vol. 14, no. 2, p. 415, 2023.

[15] W. Bao, T. Fan, G. Hu, D. Liang, and H. Li, "Detection and identification of tea leaf diseases based on AX-RetinaNet," *Sci. Rep.*, vol. 12, no. 1, p. 2183, 2022.

[16] R. Singh, N. Sharma, and R. Gupta, "Proposed CNN Model for Tea Leaf Disease Classification," in *International Conference on Applied Artificial Intelligence and Computing (ICAAIC)*, 2023, pp. 53–60.

[17] V. Jindal, V. Kukreja, S. Mehta, M. Manwal, and K. Joshi, "Emerging Paradigms: A Federated Learning CNN Approach for Tomato Leaf Disease Detection," *2023 3rd Asian Conf. Innov. Technol. ASIANCON 2023*, pp. 1–6, 2023, doi: 10.1109/ASIANCON58793.2023.10269952.

[18] V. Tanwar and S. Lamba, "Tea Leaf Diseases Classification and Detection using a Convolutional Neural Network," in *International Conference on Sustainable Computing and Smart Systems (ICSCSS)*, 2023, pp. 498–502.